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RESEARCH ARTICLE

Federated Explainable AI-Based Alzheimer's Disease Prediction With Multimodal Data

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ABSTRACT Alzheimer's Disease (AD) is a progressive neurological disease that severely impairs cognitive function. Early detection is critical for effective treatment and management. Machine Learning (ML) methods are often used to ensure early detection and prediction. However, ML has various issues, including the data island problem. The fragmentation that results from the data island problem makes building reliable, effective ML models more complex, and it is particularly problematic in industries where privacy is a concern, like healthcare. Federated Learning (FL) can help tackle the data island problem by keeping sensitive patient data decentralized and enabling many institutions to work together on model training without exchanging raw data, all while maintaining privacy compliance. As Random Forest (RF) is proven to be the best-performing classifier in this research, an RF classifier is used to create FL. The model incorporates multiple data modalities, such as Magnetic Resonance Imaging (MRI) segmentation and clinical and psychological data, to capture the variety of characteristics influencing the progress of AD. Another concerning issue with ML is its uninterpretable character. We use SHapley Additive exPlanations (SHAP) Explainable Artificial Intelligence (XAI) techniques that emphasize important factors impacting model decisions in order to improve predictability and transparency. This explainability promotes confidence in AI-based diagnoses by enabling researchers and physicians to comprehend the underlying mechanisms guiding the predictions. The combination of XAI, FL, and Open Access Series of Imaging Studies (OASIS-3) Multimodal data offers an interpretable, scalable, reliable, and privacy-centered solution for multiple complex issues, such as predicting AD. This approach results in better diagnosis precision, greater security, and increased confidence in AI technologies, making it a novel methodology in medical sciences. With data privacy maintained, our method produces 98.93% accurate predictions, providing a solid detection strategy for AD. The suggested approach's F1-score, Precision, Recall, and AUC are 98.93%, 98.94%, 98.93%, and 99.97%, respectively. This work also shows that a multimodal dataset performs better than a single modal dataset.

INDEX TERMS Federated learning, data security, data privacy, clinical data, psychological data, MRI segmentation data, explainable artificial intelligence, dementia, neurodegenerative disease.

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I. INTRODUCTION

Alzheimer's Disease (AD) is a Neuro-Digenarative Disorder (NDD) where brain cell death results in memory failure and cognitive deterioration. It is estimated that at least 50 million citizens worldwide suffer from AD and various dementias [1]. Given that Asia is the most inhabited hemisphere in the globe, a survey projects that by 2050, there will be around 71 million dementia patients in this region. About 5.5 million Americans suffer from AD, which is the most intense cause of dementia in Western societies [2]. AD does not presently have a cure. However, some drugs can reduce the illness's course [3]. The primary sign of AD is memory deterioration, which results from recent experiences. This is one of the early signals, but as the illness progresses, memory declines and novel symptoms come. A family member or close friend may be better able to identify problems if symptoms decline.

One of the most popular uses of Machine Learning (ML) in healthcare is medical diagnosis [4]. ML can identify trends in particular diseases and notify medical professionals of any anomalies. Analysis of clinical data facilitates a deeper comprehension of the fundamental processes causing illnesses. It also helps to comprehend how risk factors affect the course of a disease. There is a lot of data available to clinicians these days. It covers everything from biochemical tests to imaging device outputs and clinical symptoms. Diagnose-challenging diseases and their symptoms can be recognized and detected using ML. It also diagnoses other inherited disorders, including early-stage cancers that are challenging to detect. The issues associated with AD have gained significant attention in technical innovation and healthcare research. In this battle, ML proves to be a formidable weapon, utilizing enormous volumes of data to produce insights and prediction models that were previously unattainable. However, besides the frequently criticized "uninterpretable" character of ML algorithms, where the decision-making process is opaque and difficult to understand, using centralized datasets in ML raises severe privacy and security problems.

A primary drawback of applying ML to a centralized dataset is the heightened danger to data security and privacy. When all the data is kept in one place, it becomes more susceptible to hackers, infringements, and misuse. Large data centers may also result in inefficient processing and storage because the network needs to be able to handle heavy data loads, which can be expensive. The method may also lead to problems with compartmentalization, which restricts what can be gained from integrating datasets from various sources. The procedure can run into issues with compliance and regulations, especially if it is handling data from several jurisdictions. In addition, adversarial assaults can take advantage of model flaws and manipulate inputs to provide biased or inaccurate results. Furthermore, tracking the decision-making process can be challenging when using uninterpretable models in ML, which might impede accountability and transparency. When data is exchanged between several parties or jurisdictions, privacy concerns are exacerbated and may result in breaching data protection laws.

To address the disadvantages of centralized datasets, privacy, and security, a new approach known as Federated Learning (FL) is followed. FL allows models to be trained on decentralized datasets. The model is trained locally using the data each device (such as smartphones, Internet of Things devices, or local servers) can access. Only the updated model parameters (not the raw data) are then transmitted back to a central server for aggregation. This approach protects data since the raw data is never transferred outside the local devices. FL uses the data's decentralized nature to improve privacy and lower the dangers associated with centralized data storage. This is the main interaction between FL and decentralized datasets. FL makes collaborative model training possible without requiring the transfer of sensitive data, which is especially helpful in industries like finance or healthcare, where data confidentiality is crucial.

AD can be determined by various factors (such as clinical data, genetic and psychological information, and brain imaging). The incorporation of multiple sources of data into the diagnostic process increases the accuracy and reliability of the diagnosis. Unimodal strategies sometimes fail to capture significant information that may be present in other forms of data. Considering different modalities of data provides a more complete picture of the disease. FL enables collaborative training on such distributed datasets, combining insights from all modalities while preserving privacy.

Explainable Artificial Intelligence (XAI) is required to tackle the "uninterpretable" aspect of ML models, which renders judgments devoid of reasoning. Comprehending the rationale behind Artificial Intelligence (AI) judgments is crucial for promoting transparency, credibility, and trust in crucial fields such as healthcare, finance, and law enforcement. XAI contributes to ensuring that ML models are impartial, fair, and compliant with ethical norms in addition to being accurate. It also provides light on decision-making processes, which helps programmers debug and enhance models. XAI also makes better model openness possible, helps identify and reduce biases, enhances overall model performance, and ensures that ML models continue to reflect humans' goals and values.

This paper presents a Server-side setup for XAI in the context of FL for Alzheimer's research. This decentralized AI system aims to protect privacy by enabling cross-checking model behavior across several nodes without requiring access to specific client IDs. This methodology prioritizes an overall assessment of the FL model's explainability over client-specific insights. This work provided a unique framework designed expressly for applications in Alzheimer's research that combines the interpretability and transparency provided by XAI with the privacy-preserving advantages of FL.

Existing ML, Deep Learning (DL), and FL models for AD prediction have certain drawbacks: their performance is suboptimal when working with a single modality, data privacy regulations limit their implementation, ML creates a data island problem, and these models are not interpretable. The isolation of essential data across several businesses, institutions, or geographic areas, which hinders data exchange and collaboration, is known as the "data island problem." To mitigate the data island problem and the uninterpretable problem of ML and to ensure a highly performed model with a diverse dataset, this work proposes a privacy-preserving FL XAI approach using a multimodal dataset. Random Forest (RF) is one of the best-performing models, so this FL model is built using an RF classifier. Part of this work has been published in [5], where the superiority of multimodal datasets over single-modal data is analyzed broadly. The integration of FL, XAI, and Multimodal Processing provides a scalable, trustworthy, and privacy-preserving framework for AD prediction. It guarantees better diagnostic accuracy, enhanced security, and improved trust in AI systems, making it a cutting-edge approach to healthcare research. The main contributions of this work are listed below:

1) Proposed a novel XAI-enabled RF-FL model using a multimodal dataset to predict AD efficiently with high accuracy and ensure explainability. Due to multimodal data, the data diversity is clearly ensured, which will make the model more robust and accurate. The usage of FL and XAI made the model privacy-preserving and trustworthy.
2) Compared and analyzed the result of the proposed model with other related recent works and found that our proposed model is better than others.
3) Broadly discussed the SHapley Additive exPlanations (SHAP)-XAI for the model and found the top five features for each class and their identifying factors.

This article is structured as follows: Section II comes with the Background of AD, FL, and XAI. Related works are presented in Section III. The proposed methodology, along with the dataset description and model structure, is presented in Section IV. The result analysis is stated in Section V. Finally, Section VI comes up with the concluding remarks.

II. BACKGROUND

A. ALZHEIMER'S DISEASE

The most frequent type of dementia is AD, an advancing neurodegenerative illness that is most commonly distinguished by acute impairment of memory and cognitive deterioration. Ultimately, AD can impact conduct, communication, visuospatial direction, and motor abilities [6]. AD is thought to affect more than 6 million Americans, many of whom are 65 years of age or older. Many people experience the illness in their lives as companions or closest relatives of a person who has it. One of the early signs of AD is the inability to recall recent events or discussions. It eventually leads to severe memory loss and failure to perform daily tasks and activities. Medication may alleviate or reduce symptoms. Assistance and efforts can be used to help people with the condition and those who take care of them. AD cannot be cured. Severe brain damage at an advanced stage might lead to infection, hunger, and dehydration. There is a risk of death from these conditions.

1) SYMPTOMS

The primary sign of AD is memory loss, which results from recent discussions or experiences. This is one of the first warning signs, but as the disease progresses, memory declines and new symptoms emerge. An individual experiencing the condition may initially be aware of problems with cognition and memory. If symptoms intensify, a family member may be more likely to notice difficulties. The assessment of dementia involves two steps. First and foremost, it is critical to distinguish dementia syndromes from other illnesses that may present similarly, such as delirium, sadness, and moderate cognitive impairment. Secondly, the identification of a subtype of dementia disorder is crucial as it could dictate the appropriate course of treatment [7]. Alteration in the brain linked to AD causes increasing problems with the following elements.

- **Memory:** Everyone has memory loss from time to time, but AD-related cognition loss is permanent and gets worse with time. One's ability to function at home or work increasingly deteriorates with memory loss.
- **Thinking and Reasoning:** AD makes it hard to concentrate and think clearly, mainly while focusing on calculating concepts like numbers. It might be somewhat difficult to manage several jobs at once. It can be challenging to balance money orders, maintain an account of cash, and pay bills on schedule. Eventually, someone with AD may lose their capacity to understand and work with numbers.
- **Making Judgments and Decisions:** The capacity to reason and make sound decisions in day-to-day settings is compromised by AD. For example, someone may dress badly for the weather or make poor choices in social settings. People could find it more challenging to respond to everyday problems.
- **Organizing and Carrying Out Known Tasks:** Easy chores that need to be finished one after the other get harder. This could be cooking a meal from scratch or doing something you enjoy doing. People who have progressed AD eventually lose the ability to perform simple daily routines like dressing and bathing.

2) STAGES

The preclinical stage, also known as the presymptomatic period, can span several years or longer and is the first of the clinical phases of AD. There is no operational deterioration in day-to-day activities, minor cognitive decline, initial neurological modifications in the cerebral cortex and hippocampus, or diagnostic manifestations of AD at this stage. The moderate or initial phase of AD is characterized by several symptoms that patients first experience. These symptoms include mood fluctuations, the start of depression, trouble recalling and concentrating on things, and navigating time and location [8]. In the middle phase of AD, the disease has extended to sections of the cerebral cortex, causing more severe memory loss, which includes difficulties identifying relatives and close companions, losing control over impulses, and having issues with communication, reading, and writing. Extreme AD, also known as late-stage AD, is characterized by a progressive loss of operation and cognition, including the inability to identify loved ones, confined status, difficulty taking in and urinating, and ultimately the patient's death from these adverse effects. The disease spreads throughout the cortex, severely forming neuritic plaques and neurofibrillary tangles.

3) CAUSES

Aside from dementia, a few variables are thought to increase the risk of developing AD, even if the exact etiology of the ailment is still unknown [9]. The root cause of AD is an abnormal buildup of proteins in the brain. The buildup of tau and amyloid proteins causes brain cell death. The human brain consists of more than 100 billion neurons and other cells. Neurons collaborate to carry out all older replications required for tasks like thinking, absorbing information, recollecting, and organizing. Several factors causing AD are mentioned below:

- **Age:** The risk of developing AD rises every five years after the age of 65. Nevertheless, AD is not just a problem for older people. Twenty out of the patients with the condition are under the age of sixty-five. This kind of AD, sometimes referred to as young or early AD, can begin to affect people as early as 40.
- **Family History:** Despite the genuine rise in danger is tiny, the genes one inherits from one's parents can increase one's risk of acquiring AD. However, AD is triggered by the transfer of a single gene in a small number of families, and the disorder has a significantly greater probability of heredity.
- **Down's Syndrome:** Individuals who have Down syndrome are more likely to acquire AD. This is due to the possibility that the genetic changes causing Down's syndrome may also later lead to the build-up of amyloid plaques in the brain, which may eventually cause AD.
- **Head Injuries:** AD may be more likely to develop in those who have experienced severe head injuries, although more research is required in this area.

4) RISK FACTORS

Understanding the etiology, pathophysiology, and risk factors associated with AD remains a research focus [10]. Researchers think that a variety of risk factors, such as genes, routines, and actions, contribute to Alzheimer's and other dementias. Numerous hazards can be altered to minimize a person's chance of cognitive slump, even though other risk factors, including age and family history, may be fixed.

- **Age:** Multiple epidemiological research studies concur that age is one of the most significant risk factors for AD and cognitive deterioration among the numerous demographic characteristics, including gender, race, and social status [11].
- **Family History:** Another significant risk factor for AD is family history. Studies show that individuals with a father, brother, or sister with Alzheimer's are at an increased risk of developing the disease themselves. The risk increases if more than one family member is affected. Risk can be further increased by controllable health indicators such as sleep patterns, smoking practices, diabetes, or hypertension.
- **Sex:** The fundamental reason for this is that women usually outlive males. Nevertheless, women over 80 still have a slightly higher chance of acquiring AD compared to men of the same age. There is no certainty as to why this is happening.
- **Genetics:** It is well established that genetics contribute to AD. A person's susceptibility to an illness is influenced by two types of genes: deterministic and risky genes. Both groups have been linked to Alzheimer's genes. Deterministic genes—those that cause disease rather than raise the risk of getting one—are thought to be responsible for less than 1% of Alzheimer's cases.

B. FEDERATED LEARNING

An ML technique called FL allows building a collaborative model without centralizing the data. Important issues, including data privacy, security, and access rights, are covered by this approach, which is especially important when handling sensitive data. FL operates by requiring the following crucial steps to be completed:

1) **Initialization:** At the beginning of the procedure, a central server initializes a global model. All contributing clients' devices start with this model, which has not yet been trained.
2) **Distribution:** Every client involved has access to the original global model. The model structure is the same for all clients; however, each will train the model separately using their own or local data.
3) **Local Training:** When they receive the model, clients use their data to train it. This autonomous local training ensures that the information never leaves the client's gadget or device and protects privacy.
4) **Model Updating:** Each client updates the model following local data training. These updates may be model parameter gradients or weights depending on how FL is implemented.
5) **Aggregation:** After that, the central server receives all local model updates. The server then combines these updates to enhance the global framework or model. Federated Averaging (FedAvg), which determines the weighted average of the received updates, is one of the aggregating techniques that can be applied.
6) **Global Model Update:** The global model is updated using the combined updates. Without actually exchanging the data, this stage unifies the insights from all local datasets into a single model.

7) **Iteration:** The distribution to global model updating processes are repeated throughout multiple rounds. The global model learns from more varied local datasets with each iteration, improving its accuracy and robustness.
8) **Finalization:** The final global model is prepared for deployment and can predict fresh, unobserved data after the training is judged adequate or a specific criterion is satisfied (finishing a given number of rounds).

FL allows entities to decouple ML from the requirement to store data in the cloud by enabling collaborative learning of a shared prediction model while retaining all training data locally on the device.

C. EXPLAINABLE ARTIFICIAL INTELLIGENCE

Regarding accuracy, AI and ML have shown they can upend public services, businesses, and society. They can even outperform humans for various tasks, including speech and image recognition and language translation. DL, their most effective product in terms of accuracy, is sometimes called a ''uninterpretable'' [12]. Explaining these models gets harder and harder as they include enormous numbers of weights that carry essential information from the trained datasets. Explainable artificial intelligence, which encourages AI systems that can reveal their internal workings and explain their decision-making, was inspired by this problem [13]. XAI refers to the collection of procedures that enable general human users to understand and have faith in the output and outcomes produced by ML algorithms. Prominent figures in research, business, and government have been examining the advantages of explainability and creating algorithms that can be applied in various situations. Explainability is a prerequisite for AI clinical decision support systems in the healthcare industry, for example, according to research. This is because the capacity to understand system outputs promotes collaborative decision-making among patients and healthcare providers and offers much-needed framework accountability.

III. RELATED WORKS

Leveraging the multimodal dataset that included clinical, neuroimaging, and psychological data gathered from the Open Access Series of Imaging Studies (OASIS)-3 dataset, Jahan et al. [5] suggested a five-class AD prediction technique. To ensure multimodality, they conducted data-level fusion employing clinical data from the Alzheimer's Disease Research Center (ADRC), segmentation data from brain Magnetic Resonance Imaging (MRIs), and psychological evaluations. To choose one trait out of 39, they employed Pearson's correlation technique. They applied the Synthetic Minority Oversampling Technique (SMOTE) to build a balanced dataset because their chosen dataset doesn't have equal representation for all five classes. SMOTE was able to produce 2248 cases for each class. With 98.81% accuracy for the multimodal approach, RF was their best-performing model. For clinical, psychological, and MRI segmentation data, the obtained accuracies are 98.0%, 94.21%, and 88.85%, in that order. In addition to proposing an AD patient management system, they explainability for their work using SHAP based on features, showing which characteristic performs which significance. This work is an extension of the work of Jahan et al. [5]. The usage of multimodal datasets over a single modality was analyzed in Jahan et al. [5] this paper.

Ouyang et al. [14] have presented the ADMarker. The earliest complete solution combines novel FL algorithms with multi-modal sensors to identify multidimensional, over 20 AD digital biomarkers while protecting confidentiality. It can locate biomarkers in everyday life and create a small, multimodal device solution that can be quickly installed in homes. The framework that is being suggested is a unique three-stage FL that combines new unsupervised and weakly supervised multi-modal FL models to jointly solve many significant real-world biomarker identification issues, which comprise heterogeneity, limited computer supplies, and inadequate data tags. A four-week medical experiment was conducted on 91 senior people, of whom 31 had AD, 30 had Mild Cognitive Impairment (MCI), and 30 had Cognitively Normal (NC) conditions. ADMarker was used in this trial. With only a relatively minimal number of labeled information, the results showed the detection of over 20 everyday events with up to 93.8% detection accuracy in typical residential contexts. The scientists described their strategy as interpretable and dispersed, and after utilizing the identified digital biomarkers, they obtained 88.9% accuracy for MCI in the early detection of AD. The weak labels derived from the activity records for online supervised FL are typically brief and shallow because most people fail to document their behaviors in length. Despite its effectiveness, the system faces limitations, including high computational overhead during FL and potential challenges in handling heterogeneous data quality across institutions. These limitations highlight the need for further optimization to balance model performance with computational efficiency.

The model's effectiveness suffers when the data used to identify AD in multi-site datasets differs. Privacy concerns arise from the standard domain adaptation strategy, which includes exchanging information between origin and target regions. Federated Domain Adaptation Framework via Transformer (FedDAvT), developed by Lei et al. [15], offers an FL-based approach to this problem that may eradicate data privacy and heterogeneity. Using mean squared error for subdomain modification, the self-attention maps in the origin and destination fields are synchronized. Results have shown that using three AD datasets, their proposed model achieved accuracy rates of 88.75%, 69.51%, and 69.88% on the AD vs. NC, MCI vs. NC, and AD vs. MCI two-way classification tasks, respectively.

Bukhari et al. [16] proposed an inventive Deep Convolutional Neural Network (DCNN) model and optimized it using FL's assistance. By facilitating decentralized, multiinstitution learning and exhibiting remarkable efficiency and versatility, their model protects the confidentiality of information. With a 93% accuracy rate on the OASIS database, the model shows promise for improving patient care and AD diagnosis.

Using the FL technique, Ghosh et al. [17] have addressed the problem of data abundance for AD patients. They have used two datasets, including brain MRI pictures taken in several planes: Alzheimer's Disease Neuroimaging Initiative (ADNI) and OASIS. They gathered 436 images for the OASIS Dataset, 338 of which were regular MRI scans and 98 AD-positive. Additionally, 457 pictures were collected for the ADNI Dataset, comprising 136 AD and 321 CN MRI scans. Then, the datasets were combined for the training and testing phases to assess the model's efficiency, and then data augmentation was performed for both datasets to expand the dataset. Using the OASIS, ADNI, and combined datasets, the suggested model MobileNet underwent training in a federated manner to achieve accuracy of 95.24%, 81.94%, and 83.97%, respectively. The accuracy could be further improved using more diverse datasets and sophisticated models. There could also be room for development in the explainability of the existing work.

Almarar and Otoum [18] investigated decentralized model training for AD detection using FL in conjunction with RF and XGBoost (XGB). For this research, they also made use of the AD MRI dataset. They chose XGB and RF because of their track record of success with complicated datasets. The accuracy for RF and XGB is 94.19% and 95.53%, respectively, in the results section. FL, on the other hand, enabled numerous devices to cooperate and train this kind of ML model without centralizing the data. More direct implementation of FL could have been done to make the work more impactful.

Castro et al. [19] used the FL technique to safeguard the security and safety of the medical picture data included in the OASIS and ADNI datasets by using biometric identification for validating the images. The authors have constructed a CNN model with a 2D input layer for training reasons. There are 1052 photos in the OASIS collection, of which 526 images are evenly split between healthy and AD patients. Conversely, ADNI has an enormous archive of 6400 photos, 3200 of which are of healthy individuals and AD sufferers. Their two experiments made up their work. The CNN model was implemented in the first experiment, while the FL model was used in the second. Accuracy rates for the CNN model were 91.94% and 90.70% for ADNI and OASIS, respectively. After five learning rounds for the FL model, the maximum accuracy was achieved in Round 5, with OASIS and ADNI scoring 92.00% and 88.50%, respectively. By putting protection mechanisms in place against assaults like the model reversal attack, the reliability of the suggested system could be increased. The results may have been improved by employing more potent models, integrating the datasets, and raising accuracy.

In order to predict several classes of AD, Almohimmed et al. [20] suggested a multi-level stacking approach that integrates heterogeneous models and modalities. The ADNI cognitive sub-scores, such as the clinical dementia evaluation (sum of boxes) and the AD evaluation scale, are among the modalities. The authors trained each modality (Clinical Dementia Rating (CDR), Alzheimer's Disease Assessment Scale (ADAS), and Functional Activities Questionnaire (FQA)) in level 1 of the suggested technique using six basic models: RF, Logistic Regression (LR), Decision Tree (DT), Support Vector Machine (SVM), K-nearest Neighbors (KNN), and Native Bayes (NB). Next, construct stake testing, which combines the results of every model for the stacking training and test set and which combines the outputs of every base model for the training set. In level 2, six basic models (RF, LR, DT, SVM, KNN, and NB) are coupled in the training stack for the set used for training and the testing stack for the set to be tested to create three stacking models for each modality that is trained and analyzed. Meta-learners—RF is trained using stacking learning, and RF is assessed using stacking testing. In level 3, an additional dataset called staking training and testing is created by combining the output forecasting the stacking model from all three modalities (CDR, ADAS, and FQA) in the training and testing datasets. The meta-learner is trained using training stacking, and the last prediction is generated by evaluating the meta-learner utilizing the test set. The outcomes show that the multi-modality strategy performs better than the single-modality strategy. Furthermore, when compared to standard ML classifiers and stacking algorithms using full multi-modalities, the suggested multi-level stacking algorithms perform best with specific features. For example, they achieve accuracy, F1 score, precision, recall, and F1-scores of 92.08%, 92.01%, 92.07%, and 92.08% for two classes and 90.03%, 90.05%, 90.19%, and 90.03% for the three different classes, respectively. The performance of this model is not up to the mark. The model was adequate, but it could not adequately deal with high-dimensional data and the computational burden that comes with it. Thus, further optimizations are needed to ensure AI frameworks exhibit better scalability and generalizability on other datasets.

The BrainCrossFed model, proposed by Rashmi et al. [21], overcomes concerns about data privacy by leveraging labeled data accessible at several federated hubs and a dataset for AD to calculate a cross-fed average of model weights without transferring data to the cloud. The accuracy of the suggested DL model has grown from 99.11% to 99.77%, nearly 100%, as a result of the Cross FedAvg method. The model can also detect global minima with the help of such a cross-fed average. The suggested model achieves a final performance of 99.7% with 100%.

A Hybrid FL (HFL) framework was put forth by Baiying Lei et al. [22] to protect data privacy while simultaneously enabling the use of unlabeled data for Deep Neural Network (DNN) training. In order to depict the Region Of Interest

(ROI), the authors presented a unique Brain-Region Attention Network (BANet) that uses attention to highlight significant regions. To be more precise, the authors take the preprocessed structural Magnetic Resonance Imaging (sMRI) data and apply a brain template to extract ROI signals. Besides, we supplement the existing loss with a self-supervised loss to direct the attention map creation for learning the representations from unlabeled data. Lastly, the authors tested this approach on a multi-center database with five AD datasets. The results of the experiment demonstrate that the suggested strategy outperforms the most advanced techniques, with mean accuracy rates on the MCI vs. NC, AD vs. NC, and AD vs. MCI tests of 63.34%, 85.69%, and 698.89%, respectively. The results may have been improved by employing more potent models, integrating the datasets, and raising accuracy.

Ghosh and Gayathri [23] presented DenseFed-PSO, an FL framework combining DenseNet and Particle Swarm Optimization (PSO) for AD detection. DenseNet served as the backbone for efficient feature extraction, while PSO optimized hyperparameters to enhance model performance. The federated approach ensured privacy by training models across decentralized datasets without sharing sensitive patient information. Experimental results demonstrated superior accuracy and robustness compared to traditional methods. However, challenges include computational overhead, scalability issues with heterogeneous data, and high communication costs. Despite these limitations, DenseFed-PSO offered a promising solution for privacy-preserving, high-performance medical image analysis in Alzheimer's diagnosis.

Raisa et al. [24] developed the Cyber-Physical Fusion System aimed at stress detection using different modalities, such as physiological and behavioral data combined with social media usage. Further, a DNN was used by the system to conduct reliable stress detection by fusing information from wearable smart devices and the web. However, data integration complexity and privacy concerns were raised. The scalability of the platform, as well as privacy-preserving methods for real-world applications, were recommended for future work.

Kubi and Nazir [25] explored dementia prediction using multimodal clinical and imaging data, integrating biomarkers, cognitive assessments, and neuroimaging features to enhance prediction accuracy. By employing Gaussian Naïve Bayes (GNB) models, the study achieved a performance accuracy of 95% in detecting dementia stages, demonstrating the effectiveness of combining heterogeneous data sources. The model excels at capturing complex relationships across modalities, providing a robust and early detection mechanism. However, its limitations included high computational demands, potential biases in clinical and imaging datasets, and challenges in data harmonization across modalities. The study underscored the importance of addressing these limitations for real-world applicability and scalability.

Basnin et al. [26] proposed a novel Evolutionary Federated Learning (EFL) strategy for the diagnosis of AD utilizing the multimodal datasets available in the ADNI repository. The approach used an evolutionary model that focused on improving the effectiveness of the federated model by enhancing its capabilities in conditions where uncertainty is a key factor. For the specific case of the ADNI multimodal dataset (MRI, Positron Emission Tomography (PET), and clinical data), both baseline and EFL methods were implemented. Still, it was shown that EFL had higher accuracy in terms of AD diagnosis. The high cost of evolutionary optimization and extreme heterogeneity of the data in relation to clients served as the main limitations of this work. This work did not address the interpretability aspect of the diagnosis model. In future directions, the focus could be on developing models that are highly scalable and have high dynamic adaptability features.

According to the research done by Mitrovska et al. [27], a Secure Federated Learning (SFL) technique is suggested, which makes use of homomorphic encryption and secure aggregation techniques. They explained that this is done to conceal the data during the model training process with all the institutions involved. As assessed on the ADNI dataset, SFL preserved the patient's diagnostic details while still being able to make the diagnosis with reasonable precision. However, some of the limitations include the excessive expense of encryption and other issues that could be derived from the increase in the size of either the data set or the participants. In the future, studies could work on enhancing the efficiency required for widespread adoption and optimizing encryption techniques.

A. RESEARCH GAP

The performance of ML models in predicting AD using data from a single modality is not up to the mark, nor is there a possibility of improving the model's performance. Even ML and DL models enable data island problems. Data privacy and security are not ensured in ML and DL models. Therefore, there are possibilities to improve the model's performance using FL frameworks. Furthermore, very few existing research studies have been done using multimodal data. Still, there are scopes to work on and explore research using multimodal datasets and enhance the robustness of the model. In recent research works, various imaging and clinical data are integrated to achieve multimodality. However, Psychological data are critical in predicting AD. But, to the best of our knowledge, research works on multimodal data that contain psychological data are currently missing. Even existing FL, ML, and DL AD diagnosis models are mostly uninterpretable, meaning the general user can't know the reason behind the model's decision. Even to the best of our knowledge, there is no multimodal data-centric FL-based AD prediction model that offers interpretability using the XAI technique.

IV. PROPOSED METHODOLOGY

A. DATASET COLLECTION

For our work, we used the OASIS-3 [28]. This study incorporates information from several studies conducted at Washington University's Knight Alzheimer's Disease Research Center in St. Louis over more than 15 years. Participants are drawn from three primary cohorts: (i) cognitively general user with a CDR of 0 and a family history of AD, (ii) cognitively general user with a CDR of 0 and no family history of AD, (iii) older adults 65 years and older with a CDR of 0 (cognitively normal) or CDR 0.5–1 (very mild to mild symptoms of AD). Medical problems that interfered with long-term participation or study requirements were among the exclusion criteria. Community outreach was used to enroll participants, who then gave blood samples for genetic testing and had lumbar punctures, neuroimaging, and cognitive tests every two to three years.

OASIS-3 consists of structural and functional magnetic resonance imaging, amyloid and metabolic PET imaging, neuropsychological testing, and clinical data from 1,098 individuals, of whom 605 are cognitively normal adults, and 493 are at various stages of cognitive decline. These participants range from 42 to 95. There, an identification was given to the data of each participant, and all dates were removed and uniformed to represent the number of days since their membership. A large number of Magnetic Resonance (MR) Volumetric segmentation files produced by Freesurfer accompanied the sessions. Due to these psychological data, Freesurfer volumetric MRI segmentation data and ADRC clinical data are utilized in this research. Table 1 lists all the data for each modality together with several characteristics.

1) ADRC CLINICAL DATA

In compliance with the National Alzheimer Coordinating Center Uniform Data Set (UDS), participants completed clinical evaluation protocols. The UDS examinations included medical history, neurological evaluation, physical examination, and family history of AD. Clinical and cognitive assessments were performed every three years for those ages 64 and under and annually for those ages 65 and older. The ADRC clinical data [27] includes longitudinal data from 1098 unique patients collected over time. Longitudinal data is data that is consistently gathered periodically from the same participants. The numbers for participants of CN, AD, Uncertain, Non-AD, and Other classes are 4476, 1058, 505, 142, and 43, respectively. The classes such as CN, AD, unclear dementia, and non-AD dementia all have unique characteristics. Non-AD dementia labels include vascular dementia, Parkinson's disease, Dementia with Lewy Bodies Disease (DLBD), and frontotemporal dementia. The dataset's key features included age, judgment, Mini-Mental State Exam (MMSE), personal care, APOE (apolipoprotein E gene), memory, height, CDR, weight, Orient (latest and long-term memory tests), and Sumbox. Each year, the ADRC offers a neuropsychological test for people aged 65 and upwards. The Mini-Mental State Examination (MMSE) measures overall cognitive ability, including scores spanning 0 (severe impairment) to 30 (with no impairment).

Subjects underwent clinical evaluation techniques in line with the National Alzheimer Coordinating Center's UDS. UDS evaluations included physical tests, historical health events, and a neurological assessment. Every three years, subjects aged 64 and below receive clinical and psychological evaluations. Subjects aged 65 and up underwent annual clinical and cognitive examinations. The UDS employed the CDR Scale to evaluate participants' dementia position. A participant was deemed to have moderate dementia if their CDR score was 2, but when they came to CDR 0.5, they were deemed no more suitable for in-person evaluations. The OASIS data "ADRC Clinical Data" (UDS form B1 and B4 variables) includes the following information: weight, height, age at admission, and CDR assessments. As a component of the evaluation, clinicians conducted a diagnostic impression input and discussion. This led to a tagged dementia diagnosis that was documented in the OASIS data "ADRC Clinical Data." Diagnoses for this variable include "AD dementia," "regular cognition," "vascular dementia," and vitamin deficiencies, as well as drunkenness and disorders of mood. While there may be some overlap in the diagnostic results, the determination of the factors dx1–dx5 is different from UDS assessments.

2) PSYCHOLOGICAL DATA

Numerous well-known psychological tests, including the Boston Naming Test, animals, Trailmaking B (Trail B), Trailmaking A (Trail A), vegetables, digit span, logical memory, digit symbol, Wechsler Adult Intelligence Scale (WAIS), and others, are included in the psychological evaluation dataset [28]. Digit Span involves participants repeating a series of digits in both directions in order to assess their memory and attention range. The participant's score was derived from the number of trials that were successfully repeated both backward and forward, in addition to the most extended duration that the subject could repeat backward.

Semantic assessments of memory and language include the Boston Naming Test, which asks participants to name pictures of everyday things, and the Category Fluency Test, which asks participants to name as many words as possible that fall into a category, including "vegetable" and "animal." Psychomotor speed was measured using the Trail Making exam, Part A, and the WAIS-R Digit Symbol exam. The WAIS-R Digit Symbol test is scored based on how many pairs of digit symbols are completed in ninety seconds. The executive-level function test was administered using the Trail Making Test Part B. In part A of the Trail Making Test, subjects were required to link a series of digits (1)-26) and, in part B, a series of alternating numbers and letters (1)-A-2-B) in order to construct a trail. The number of commission mistakes, the number of proper lines, and the total time to completion in seconds—with a maximum of 150s for Trails A as well as 300s for Trails B—are every measure of outcome.

TABLE 1. A brief explanation of multimodal dataset [5].

Data Modality	Unique Participants	Longitudinal Data	Number of Features
Clinical Data	1098	6224	13
MRI segmentation Data	1053	2047	10
Psychological Assessment data	810	3342	17

The Wechsler Memory Scale–Logical Revised Memory—Story A assesses episodic memory. Subjects are asked to recall facts from a brief narrative with 25 bits of data after an expert reads it loudly and again following a 30-minute lag. Scores range from 0 (no memory) to 25 (total recall).

3) MRI SCAN SEGMENTATION DATA

The Knight Alzheimer Research Imaging Program at Washington University in St. Louis performed all of the neuroimaging studies. Three distinct Siemens scanner models—Vision 1.5T, TIM Trio 3T (two separate scanners of this kind), and BioGraph mMR PET-MR 3T—were used to capture MRI data (Siemens Medical Solutions USA, Inc.). For archival purposes, every session concurrently recorded on the Biograph mMR scanner has been divided into separate PET and MRI sessions. In these sessions, the images were analyzed by an open-source software suite called Freesurfer, which can process and interpret MRI scans of an individual's brain. This Freesurfer dataset provides volumetric data for various human brain regions, including the total cortex, the intracranial, hemisphere cortex, supratentorial, subcortical shades of gray, overall gray, and hemisphere cortical white matter [28].

The Extensible Neuroimaging Archive Toolkit (XNAT) pipeline of Freesurfer was used to perform the image analysis. FreeSurfer v5.0 or v5.1 has been used to handle T1-weighted pictures. The technique encompasses motion alteration and volumetric T1 weighted image averaging, computerized Talairach change, tessellation of the gray matter-white matter limit, magnitude normalization, computerized topology, division of subcortical white matter and deep gray matter dimensional structures (including the hippocampus, putamen, caudate, amygdala, and ventricles), and removal of non-brain tissue using an integrated exterior extending technique. After completing cortical models, flexible processes were used for data processing.

After cortical algorithms were finished, several deformable rules were carried out for additional data processing and analysis. These procedures included surface increment, parcellation of the cortex into units according to gyral and sulcal framework, authorization to a spherical atlas depending on specific cortical bending structures to correlate cortical geometry throughout subjects, and generating various kinds of surface-based information, such as corresponding curvature and sulcal depth. This approach is used in division and deformation processes to create illustrations of cortical thickness, which is computed as the closest distance between the white/gray limit and the cerebrospinal fluid (CSF)/gray boundary region at every vertex within the tessellated surface area, using continuous information from the whole three-dimensional MR volume. The generated maps can identify submillimeter variances across groups and are not limited by the information's voxel resolution. Procedures for measuring cortical thickness have been verified through comparison with manual measurements and histological investigation. Freesurfer morpho metric techniques have proven reliable in test-retesting across scanner brands and field strengths.

B. PRE-PROCESSING OF MULTIMODAL DATASET

This research project aims to use multimodal data to predict AD. Here, multimodal data is created through data-level fusion. To achieve multimodality, we have combined three distinct domain datasets: MRI (more accurately, brain MRI) segmentation and clinical and psychological data. Three separate datasets are integrated at the start of the data fusion approach. The OASIS website has undergone this integration. This website offers the ability to combine various patient domain data according to subject ID and session.

1) QUANTITY OF EACH MODALITY'S INSTANCES

3342 occurrences of the same participants were found in the Psychological and Clinical assessments once the combined dataset was created. That being said, there are 3220 cases in the MRI segmentation dataset. Thus, there are 122 missing instances in the MRI segmentation dataset (3342-3220) = 122. There were 810 (clinical and psychological) and 799 (MRI segmentation) distinct subjects. The combined dataset has 34 different kinds of labels. All 34 types of labels fall into five main categories: uncertain dementia, CN, AD, non-AD, and others.

TABLE 2. RMSE value after data imputation using KNN [5].

Neighbors Number	RMSE
2 Neighbors	**0.806**
3 Neighbors	0.834
5 Neighbors	0.904

2) KNN-BASED DATA IMPUTATION

At this point, the missing values are filled in by applying KNN imputation. The average value of every sample's missing values is calculated using the N nearest neighbors of the training group. If the features of two samples are similar and none is absent, then they are near. The root mean square error (RMSE) of the imputed collection with varying numbers of neighbors is displayed in Table 2. Two neighbors have a lower RMSE score than the others. Consequently, two neighbors are imputed to the joined or fused dataset.

3) SELECTION OF FEATURES

The multimodal dataset has 39 features, but not every feature will have the same impact on the prediction. Both single datasets and multimodal datasets undergo feature selection operations to determine the key features. The Pearson's Correlation is used to choose features. A linear relationship between two quantitative variables is ascertained using Pearson's correlation, shown in Figure 1. It is regarded as the most effective method for quantifying the relationship between variables because it depends on the covariance technique. It shows the trend of the association as well as the magnitude of the correlation.

Additionally, the Boruta feature selection approach and cross-validated Recursive Feature Elimination (RFE) were utilized initially to pick significant features. However, it turns out that each of these two strategies has disadvantages of its own. The total amount of features used for ML models varies after every RFE execution, and RFE is computationally costly. As RFE was not a stable feature selection approach for this work, therefore it was not an appropriate feature selection strategy for this investigation. Additionally, Boruta did not offer a high accuracy score. In this study, Pearson's correlation is favored based on these disadvantages.

For Pearson correlation mapping, the range of values is -1.0 to 1.0. 1.0 denotes perfect correlation, while -1.0 denotes no correlation. It is not necessary to focus on two features that have a significant correlation when selecting features. To decrease errors and reduce computing complexity, one trait between these two should be chosen. For the Clinical dataset, 0.9 is the threshold value that performs the best. The optimal value for the MRI dataset is obtained within the thresholds of 0.9 and 0.95. The most promising results for the Psychological dataset are obtained at threshold values of 0.75, 0.8, and 0.9. Using the Psychological dataset, threshold readings of 0.75, 0.8, and 0.9 yields the most promising findings.

For multimodal datasets, CortexVol, RhCortexVol, CorticalWhiteMatterVol, TotalGrayVol, and RhCorticalWhiteMatterVol were discovered to be the least influential feature with strong correlation values (features with robust correlations should be eliminated since they lead to volatile models, numerical mistakes, and subpar prediction accuracy) at the threshold of 0.95 Pearson correlation. For this work, the selected features are: DIGIF, LOGIMEM, DIGIB, DIGIBLEN, TRAILARR, ANIMALS, TRAILA, TRAILBRR, MEMUNITS, TRAILALI, TRAILB, TRAILBLI, WAIS, MEMTIME, BOSTON, lhCortexVol, IntraCranialVol, SupraTentorialVol, DIGIFLEN, LhCorticalWhiteMatterVol, AgeAtEntry, Homehobb, Judgment, Commun, Memory, VEG, SubCortGrayVol, Orient, Perscare, MMSE, APOE, Sumbox, Height, and Weight.

4) DATASET BALANCING

The multimodal dataset was imbalanced, meaning the number of instances in each class was unequal. The CN class contained 2248 data instances, whereas AD contained 669 samples. Non-AD, Uncertain dementia, and Others contained 106, 287, and 26 cases, respectively. For making an imbalanced medical dataset balanced, data oversampling is a solution. Here, oversampling using the SMOTE is employed to create a balanced dataset. SMOTE is one of the most widely used oversampling strategies to deal with the imbalance problem. It aims to correct the class distribution by randomly selecting minority class samples and replicating them. By merging minority instances that already exist, SMOTE generates new minority instances. Following oversampling, it was discovered that there were 2248 cases in each of the five classes. Therefore, after oversampling, the total number of cases became 11240, mentioned in Table 3.

C. MODEL IMPLEMENTATION

1) FEDERATED LEARNING MODEL

An ML technique called FL allows building a collaborative model without centralizing the data. This architecture can be better described in four steps:

Initialization: Firstly, the central server has initialized the global model. The global parameters or meta parameters of the model are set by the central server known as the coordinator (e.g., weights in a ML model). These parameters are set to a pre-defined range of values, which is zero. An upper configuration is provided for a fixed set of hyperparameters to be set, such as the learning rate, number of iterations (rounds), and the approach to obtain the model (e.g., FedAvg). All clients have forwarded the initial model, which has been trained using the global data, to all devices that are part of it.

Local Training: Secondly, all the client devices have started executing using this RF model. Here, the number of clients is kept at 10 because this setup ensures the best result. Each client trains this RF model using its dataset separately. This is local training, and after this process, the model parameter gradients or weights are transferred to the global model.

Aggregation: Thirdly, using these updated weights, the central global model will train itself, improving its accuracy and other performance parameters. The server uses the FedAvg technique to combine these updates to enhance the global framework or model.

Global Update: Fourthly, the global model will pass the updated RF to each local model, and the local models will again train RF with their dataset. Finally, this process will continue for 50 epochs as per this work, and every time, clients will train RF and pass the updated weights.

2) ALGORITHM

The FedAvg algorithm (Global Model) is written below in Algorithm 1. This algorithm can be described in 8 steps.

1. Setting up the Model *Vo*:

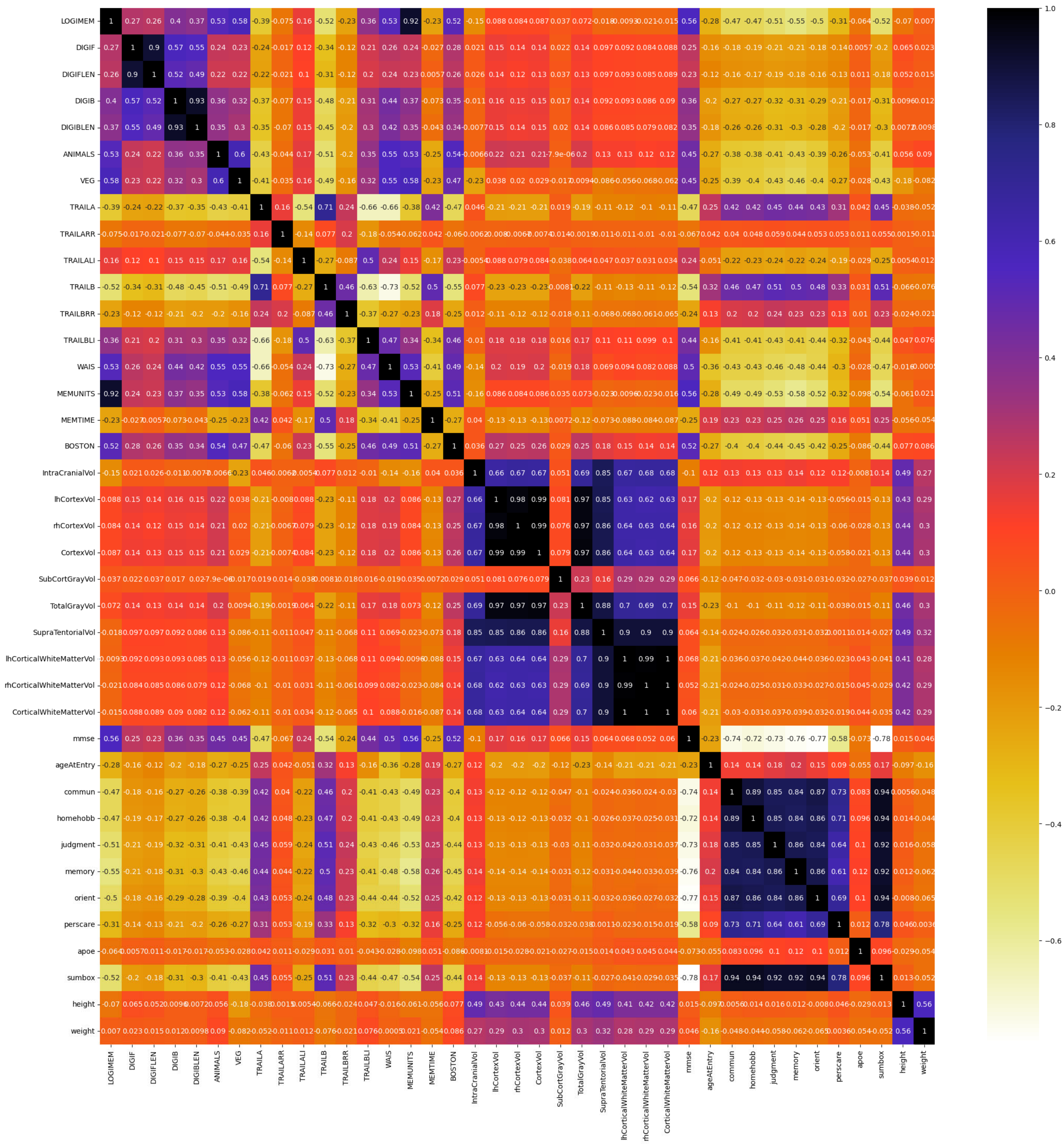

FIGURE 1. Pearson's heatmap with 39 features of multimodal dataset for correlation [5].

TABLE 3. Five classes and their number of samples [5].

Class	Number of Data Before Balancing	Number of Data After Balancing
CN	2248	2248
AD	669	2248
Non-AD	106	2248
Uncertain Dementia	287	2248
Others	26	2248

The first step in the procedure is to initialize the global model Vo. It contains all of the model's parameters, including weights and biases.

2. Iterative Training Rounds:

Here, the algorithm operates in a series of communication rounds ($t = 1, 2, \ldots$), during which information is periodically exchanged between the central server and the participating devices.

3. Client Selection (Rt):

In each round, a subset of the available clients (Rt) is picked. A random set of max (C,P,1) was defined as Rt. C represents a portion of the clients who participate, and P represents the total number of devices. This helps to balance participation variety and communication expenses.

4. Broadcasting the Global Model:

The central server provides the most recent global model $Vt - 1$ to all selected devices Rt. During the training session, all devices are trained using the same model.

5. Local Device Training:

The selected device (P) uses the local dataset to train the global model $Vt - 1$. The devices adjust the parameters of the local model to obtain Vt^P based on local data.

6. Local Updates Are Being Uploaded:

After local training, the devices provide updated model parameters to the central server Vt^P. These updates are usually limited to model weights or gradients, ensuring that no raw data is communicated and data privacy is protected.

7. Global Aggregation:

On the central server, all the updates from all the devices that participated in the training Vt^P are merged to produce a new version of the global model Vt. This process makes use of the FedAvg technique. For every device, all updates are weighted proportionately with respect to the size of the data set available on the device.

8. Iterative Process:

At the conclusion of each training round, the global model is modified Vt and sent back to all the selected devices to begin the subsequent training round. This process is repeated for a set number of rounds or until the model reaches the desired target effectiveness.

Algorithm 1 Algorithm of FedAvg (Global Model)

Input: Vo

for each round t = 1,2,..., do Rt=(random set of max (C,P,1) devices);

for each device P belongs to Rt in parallel do

Broadcast recent global model $Vt - 1$ to device P;

Receive the update from the local model Vt^P from the device P;

Update the global model Vt.

The FedAvg algorithm's local device component describes how each device (or client) adds to the collaborative learning process in an FL system. The provided Algorithm 2 is explained in detail below:

1. Initial Process:

This is an iterative process, which means that the algorithm proceeds in sync with the global server for multiple rounds. Each round is a cycle in which the local device receives, trains, and updates the global model.

2. Receiving the Updated Global Model:

At the start of every round, the local device is provided with the updated global model V_t from the central server. This enables all the devices to commence their local training from the identical state of the global model.

3. Local Training Using the Updated Global Model:

Each local device applies the received global model as the initial state of its regional training. The model is trained on the global model using the device's private, locally stored data. RF is applied to optimize the model on the local dataset.

4. Computing the Updated Local Model:

Upon completing the training process on the local data, the device calculates its updated local model parameters ($V_t P$). These updates incorporate the information gained from the local data. Still, the changes are made in a way that keeps the information secure, meaning that the original files are never disclosed.

5. Disseminating the Updated Local Model:

The relevant updates to the local model parameters are returned to the central server for further storage and computation. Communication is limited to exchanging model updates, such as weights, update gradients, and model parameters.

Algorithm 2 Algorithm of FedAvg (Local Device)

for each round t = 1,2,..., do;

Receive the updated global model;

Use the updated global model to train the local model;

Compute the new updated local model Vt^P;

Share the recently updated local model to global model;

3) RANDOM FOREST GLOBAL MODEL DESCRIPTION

For implementing the global model of the FL, the RF model gives the best performance. The specifics of its implementation are covered here. In this instance, the basis estimator is the decision tree classifier. Each estimator is trained using a different bootstrap sample that is extracted from the training set. Estimators use all attributes for both training and prediction. The forest contains one hundred trees. A statistic called Gini is used to evaluate a split's quality. For an internal node to be divided, at least two samples are needed. Up until each leaf has fewer than a pair of samples, nodes are increased. There must be a single sample present at minimum for a leaf node to exist. The square root function is used to determine how many attributes to look at while figuring out the best split. The tree can reach a maximum level or depth of 30 and a minimum of one node, which can function as a leaf. The tree is grown using the Best-First method. The relative reduction of impurities distinguishes the most effective nodes. Bootstrap samples are employed in tree construction. Every time the model is executed, the identical test and train sets are given to it because the random state is maintained at 42. All five classes are supposed to be allocated weight one.

4) EXPLAINABLE AI MODEL

A set of frameworks and tools known as "XAI" is intended to help people comprehend and analyze the predictions generated by ML models. It also helps people understand the model's behaviors and enables debugging and performance improvements. Any ML model's output can be explained using the game-theoretic method known as SHAP. This method uses the standard Shapley values and related expansions from game theory to connect optimal credit distribution to local explanations. A feature value's Shapley value is its impact on the payment, weighted and aggregated across all possible feature value combinations. The SHAP value for a feature k is calculated as:

$$\phi_k(val) = \sum_{G \subseteq \{1,\ldots,h\} \setminus \{k\}} \frac{|G|!\,(h-|G|-1)!}{h!} (val\,(G \cup \{k\}) - val(G)) \tag{1}$$

Here, in equation 1, $\phi_k(val)$ is the SHAP value for feature k, representing its contribution to the prediction. h denotes the set of features. G is the subset of features excluding i. $val(G)$ determines the model's prediction when only the features in subset G are considered (other features are treated as missing).

$$val_x(G) = \int \hat{z}(w_1, \ldots, w_h) d\mathbb{H}_{w \notin G} - E_W(\hat{f}(W)) \tag{2}$$

In equation 2, $val_x(G)$ represents the value or contribution of the subset G of features to the model's prediction for a specific input x. $\hat{z}(w_1, \ldots, w_h)$ describes a modified function or approximation of the model output $\hat{f}$, evaluated with some or all features from the subset G. $d\mathbb{H}_{w \notin G}$ refers to the marginal distribution over the features not included in G. Integration over this distribution accounts for the uncertainty or average contribution of the excluded features. $E_W(\hat{f}(W))$ refers to the expected value of the model's output $\hat{f}$, averaged over the entire distribution $H(W)$ of all features. This is the baseline prediction, representing the model's behavior without specific feature contributions

$$\begin{aligned} val_w(G) &= val_w(\{1,3\}) \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \hat{z}(w_1, W_2, w_3, W_4) d\mathbb{H}_{W_2 W_4} - E_W(\hat{f}(W)) \end{aligned} \tag{3}$$

For each missing feature G, several integrations must be executed. Equation 3 provides an example: applying a model with four features (w1, w2, w3, and w4) and estimating the coalition G based on feature values w1 and w3.

This is comparable to the feature contributions of the linear model. The only attribution method that satisfies the criteria of Efficiency, Symmetry, and Additivity—all of which are useful in defining a just compensation—is the Shapley value. Therefore, the contribution of a variable (or several variables) to the difference between the model's predicted value and the average of all individual predictions is the Shapley value for a given individual. To make this happen:

Step 1: Calculating the Shapley values for a particular subject. For the input characteristics, simulate different combinations of values.

Step 2: Calculate how much each combination's average forecast differs from the predicted value. Consequently, a variable's Shapley value is equal to the average of the value contributions resulting from the different combinations. Tree SHAP is a quick and precise method for computing SHAP values for tree models and ensembles under various feature dependence assumptions.

The SHAP Tree Explainer is used in this research endeavor to analyze and see all of the decision-making aspects and their percentages. This method makes it simple for a physician or patient to comprehend the model's reasoning, as well as the characteristics and proportions that go into the decision-making process.

D. PROPOSED MODEL ARCHITECTURE

There will be a server-side global model that will work on the RF model. Initially, this model will be transferred to all the clients. Clients will train the model using their local data and pass the parameters and model weight to the global model. Using the FedAvg technique, all the updated weights will be accumulated into the global model. The newly updated model will be passed on to the clients again, and this process will continue until all 50 epochs are completed. Here, in the global model, there will be a SHAP explainer that will interpret the uninterpretable RF global model as a white box RF global model. Using this XAI model, the central administrator can identify whether the model is performing correctly as well as which features are the most impactful for classifying each class. As FL supports data privacy, this central administrator XAI will not breach individual patients' data privacy. Figure 2 illustrates the proposed Federated XAI-based AD Prediction Model Architecture.

V. PERFORMANCE ANALYSIS

A. PERFORMANCE METRICS

An evaluation measure describes a model's performance. The confusion matrix stands out across numerous metrics. A confusion matrix generates a matrix that summarizes the overall effectiveness of the model. The matrix is $N \times N$, with N being the total number of classes anticipated. There are four essential terms for calculating the confusion matrix: True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN). Some crucial definitions connected to the confusion matrix are:

1) **Accuracy:** Accuracy is the most widely used parameter for evaluating classification models. This statistic calculates the fraction of correct predictions among all cases evaluated. The best use case occurs when the

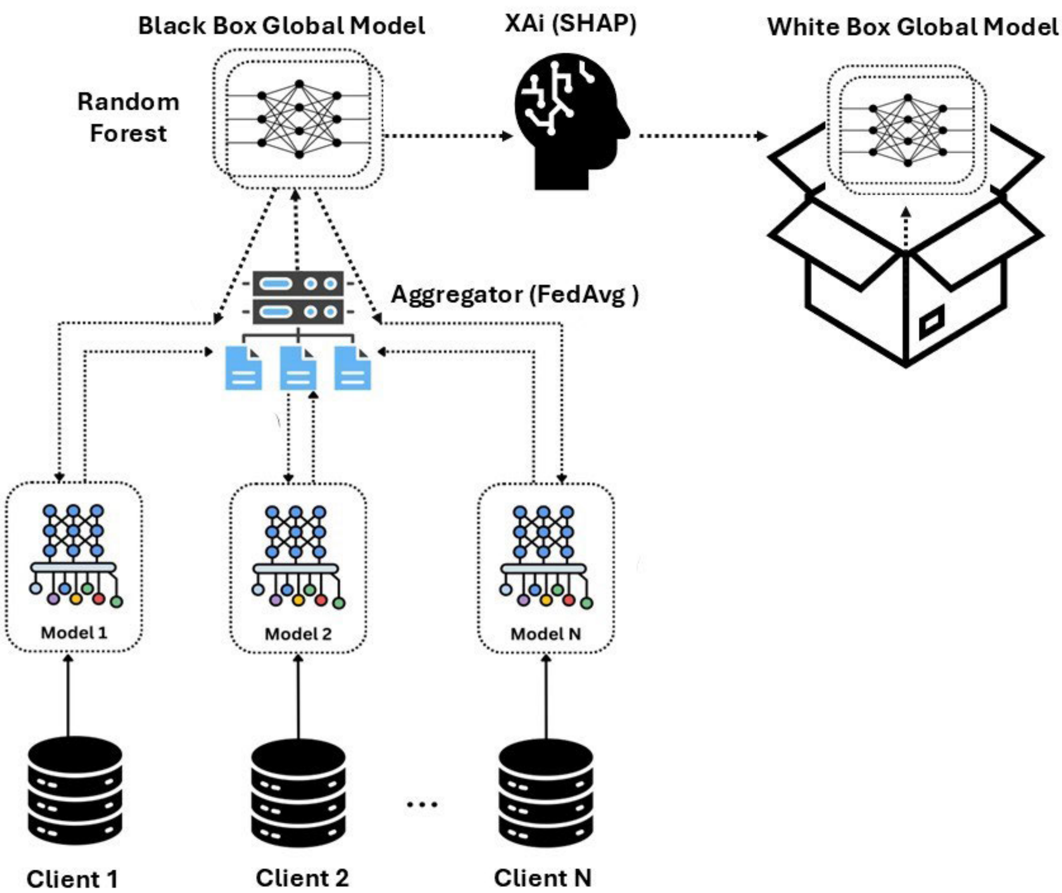

FIGURE 2. Federated explainable AI-based Alzheimer's disease prediction framework.

data is balanced. Equation 4 describes the method of calculating Accuracy.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

2) **Precision:** From the overall predicted patterns, the measure of precision is used to calculate how well the positive patterns in the positive classes were anticipated. It is the proportion of successfully identified positive cases. Equation 5 describes the method of calculating Precision.

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

3) **Recall or Sensitivity:** Recall is the total number of actual positive cases that were accurately forecasted. This statistic is also known as sensitivity. Equation 6 describes the method for calculating Recall.

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

4) **F1-Score:** The F1 score can be understood as a harmonic mean of precision and recall, with 1 representing the best score and 0 representing the lowest score. Equation 7 describes the method of calculating F1-Score.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (7)$$

5) **AUC:** The Area Under Coverage (AUC) score represents the area under the region of the coverage curve. It summarizes a model's ability to provide relative scores that distinguish between positive and negative examples across all categorization thresholds. The AUC score ranges from 0 to 1, with 0.5 representing random guessing and 1 indicating perfect performance.

B. PERFORMANCE ANALYSIS WITH THREE INDIVIDUAL DATASETS

Table 4 shows the individual dataset performance for different models. For feature selection in psychological, clinical, and MRI segmentation datasets, the 0.9 thresholds produced the maximum accuracy. For all three datasets, the RF model outperforms the other models (DT, LR, KNN, Multilayer Perceptron (MLP), Ada Boost (AB), SVM, and XGB) in terms of accuracy. All of these results were achieved using a five-class classification. Here, the used features of Psychological dataset: DigiB, DigiF, Logimem, Animals,

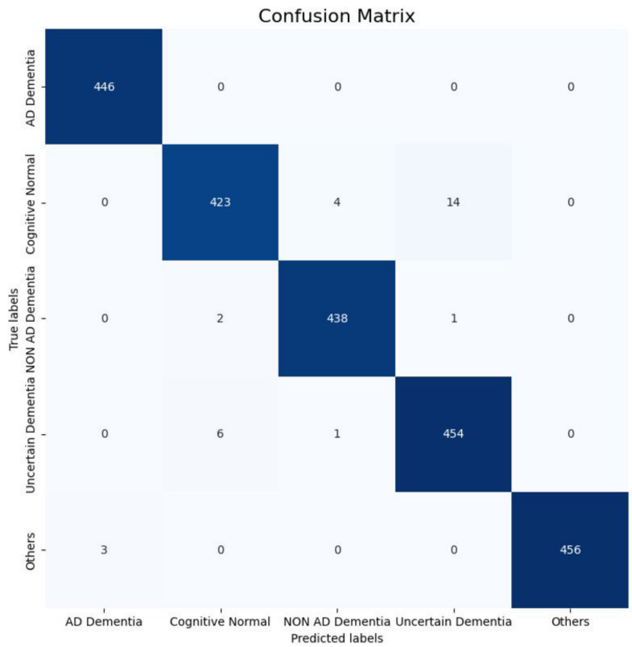

FIGURE 3. Confusion matrix of the random forest classifier using multi-modal dataset.

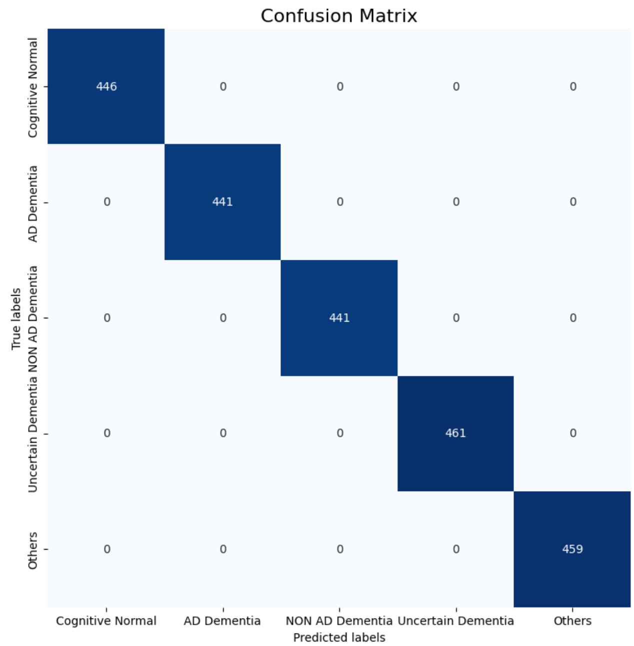

FIGURE 4. Confusion matrix of the random forest federated learning model using multi-modal dataset.

Veg, TrailBrr, TrailA, TrailAli, TrailArr, TrailB, TrailBli, WAIS, Memtime, and Boston. MRI Segmentation dataset: IntraCranialVol, lhCortexVol, SubCortGrayVol, SupraTentorialVol, and LhCorticalWhiteMatterVol. Clinical dataset: Mmse, AgeatEntry, Judgment, Commun, Perscare, Memory, Homehobb, Orient, Apoe, Height, and Weight.

C. ANALYZING PERFORMANCE WITH THE MULTIMODAL DATASET

Table 5 makes it evident that, when using the multimodal dataset and a feature selection threshold of 0.95, RF is the best-performing algorithm for five-class prediction. This indicates that a feature is ignored if its correlation value is more significant than 0.95. Animals, Logimem, Digif, Digib, DigiBlen, TrailBli, Veg, DigiFlen, WAIS, Memunits, Traila, Trailarr, Trailbrr, Memtime, Boston, Trailali, Trailb and so on are the characteristics that are employed in this instance. Apoe, Height, Weight, Commun, Memory, Orient, Homehobb, Judgment, Persecare, Apoe, lhCortexVol, LhCorticalWhiteMatterVol, SubCortGrayVol, SupraTentorialVol, Mmse, Age At Entry, and Sumbox.

Table 5 describes the stratified 10-fold cross-validation, testing, and training accuracy. The multimodal data has an improved accuracy rating (10-fold cross-validation) compared to the individual dataset. In this case, the cross-validation accuracy is employed since it prevents a model from being overfit. The RF classifier's precision, recall, and F1-score can be calculated based on the confusion matrix (Figure 3). The RF model's training, testing, cross-validation accuracy, F1-score, precision, AUC, and recall values are 100%, 98.84%, 98.81%, 98.75%, 98.94%, 99.97%, and 98.79%, respectively. RF's ability to handle complex, multimodal, and noisy datasets with nonlinear relationships makes it ideal for AD prediction. It leverages ensemble learning to reduce overfitting and improve generalization, leading to better performance than other models.

D. MULTIMODAL DATASET WITH OR WITHOUT FL APPROACH

From Table 6, it is clear that the performance of the FL approach is higher than the without FL approach. The testing accuracy of FL is 98.93%, which is higher than with our FL approach. The precision, recall, F1-score, and AUC value are 98.93%, 98.94%, 98.93%, and 99.97%. Due to data diversity, minimal overfitting, scalability, model robustness, and maintaining data heterogeneity, the FL approach outperforms the without-FL approach. Figure 4 shows the confusion matrix of the FL approach.

E. COMPARISON OF MODELS WITH EXISTING WORKS

It is observed from Table 7 that the accuracy of the ADmarker model [14] for predicting the biomarker of AD is 88.9%. That of the FedDAvt [15] model is 88.57% in predicting AD. Ghosh and Gayathri [23], Castro et al. [19], and Bukhari at el. [16] used the FL technique using an unimodal dataset, achieving the performances of 95.24%, 92%, and 97%, respectively. Baglat et al. [29], and Kavitha et al. [30], and Almarar and Otoum [18] used unimodal (MRI) OASIS data, RF model as a classifier, and ensured an accuracy of 86.8%, 86.92%, and 94.19%, respectively. Furthermore, Amrutesh et al. [31] used Text data from OASIS and RF model for classification and secured 92.13% accuracy. On the other

TABLE 4. 10-fold cross-validation accuracy of various ML models using individual datasets [5].

Dataset	RF	DT	LR	KNN	MLP	GB	AB	SVM	XGB
Clinical	**98%**	95.0%	48.6%	88.8%	70.2%	48.4%	48.4%	45.1%	89.0%
MRI	**88.85%**	85.4%	32.8%	82.0%	21.6%	71.9%	44.2%	24.5%	67.2%
Psychological	**94.21%**	81.4%	43.0%	85.4%	54.6%	82.2%	55.9%	26.7%	77.2%

TABLE 5. Analyzing accuracy using the multimodal dataset [5].

Performance Metrics	RF	DT	LR	KNN	MLP	GB	AB	SVM
Training Accuracy	**100%**	100%	30.5%	89.1%	25.8%	97.7%	54.6%	29.5%
Testing Accuracy	**98.84%**	94.2%	31.6%	82.2%	25.4%	96.3%	54.4%	29.5%
Cross-Validation Accuracy	**98.81%**	94.9%	31.3%	83.8%	25.8%	95.6%	55.6%	25%
Precision	**98.94%**	94.5%	30.6%	82.5%	46.6%	96.3%	52.2%	24.1%
Recall	**98.79%**	94.4%	31.8%	82.2%	32%	96.2%	54.6%	21.9%
F1-score	**98.75%**	99.1%	28.7%	81.9%	29%	96.2%	47.5%	12.2%
AOU	**99.97%**	96.5%	63%	95.8%	53.4%	99.7%	79%	58.3%

TABLE 6. Performance comparison of multimodal dataset with FL approach versus without FL approach.

Performance Metrics	FL approach with Multimodal Dataset	Multimodal Dataset
Training Accuracy	100%	100%
Testing Accuracy	**98.93%**	98.84%
Precision	**98.93%**	98.75%
Recall	**98.94%**	98.94%
F1-Score	**98.93%**	98.75%
AUC	**99.97%**	99.97

TABLE 7. Comparison of the proposed work with other recent related works.

Ref.	Year	Model	Dataset	Accuracy
Baglat et al. [29]	2020	RF	MRI data from OASIS	86.8%
Kavitha et al. [30]	2022	RF	MRI data from OASIS	86.92%
Amrutesh et al. [31]	2022	RF	Text values from OASIS	92.13%
Lei et al. [15]	2023	FedDAvT	-	88.75%
Ghosh et al. [17]	2023	FL	MRI data from OASIS	95.24%
Almohimmed et al. [20]	2023	RF	ADNI	92.08%
Ouyang et al. [14]	2024	ADmarker	-	88.9%
Castro et al. [19]	2024	FL	OASIS	92%
Bukhari et al. [16]	2024	CNN-FL	MRI data from OASIS	97%
Almarar et al. [18]	2024	RF	OASIS	94.19%
Kubi et al. [25]	2024	GNB	Multimodal OASIS (Clinical and Imaging)	95%
The Proposed Work	**2025**	**RF-FL**	**Multimodal-OASIS (Clinical, MRI Segmentation, and Psychological)**	**98.93%**

hand, Kubi and Nazir [25] proposed a GNB model for predicting dementia using clinical and imaging data from the OASIS repository and achieved 95% accuracy. The details of the internal model of these related works are mentioned in Section III. In contrast to all these aforementioned works, this work proposed an RF-FL approach using Multimodal data from the OASIS repository, which is why this work has achieved better performance (accuracy 98.93%) than other existing models. Figure 4 shows the confusion matrix of the RF-FL model.

F. EXPLAINABLE ARTIFICIAL INTELLIGENCE

A SHAP violin plot shows how each feature affects model predictions. It combines the qualities of a violin plot and an overview plot, displaying both the distribution of SHAP values and the size of each feature's influence on the result of the model. Here are ways to read essential elements of the SHAP violin plot. The model's features are presented along the y-axis in descending order of relevance. Each feature correlates to one violin plot, which depicts how it influences model predictions. The x-axis depicts SHAP values, which indicate how each feature affects the model's output. Positive SHAP values indicate that a characteristic improves the projected outcome, whereas negative SHAP scores indicate the opposite. The greater the feature's influence on the prediction, the more SHAP value deviates from zero.

The width of each violin plot represents the distribution of SHAP values for that feature over every dataset record. A broader view of the violin plot reveals that more cases had SHAP values of that magnitude. Narrow sections represent fewer instances of specific SHAP values. The figure's symmetry indicates both the characteristic's positive and negative effects. The violin plot's point colors, which typically range from blue (low values) to red (high values), represent the

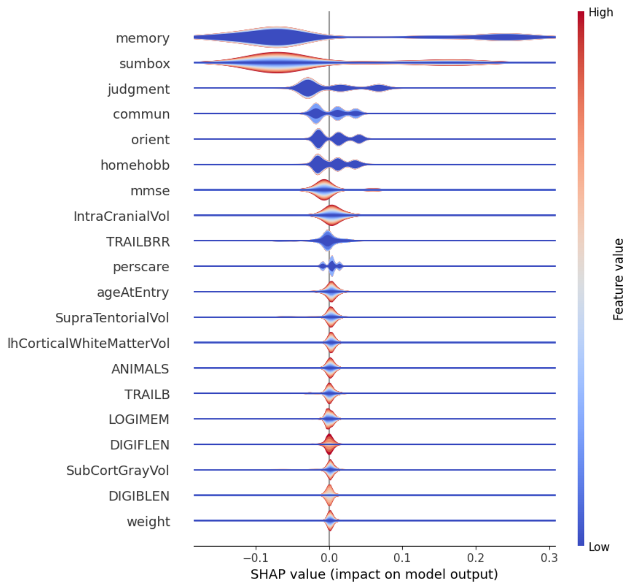

FIGURE 5. **SHAP violin plot for CN class.**

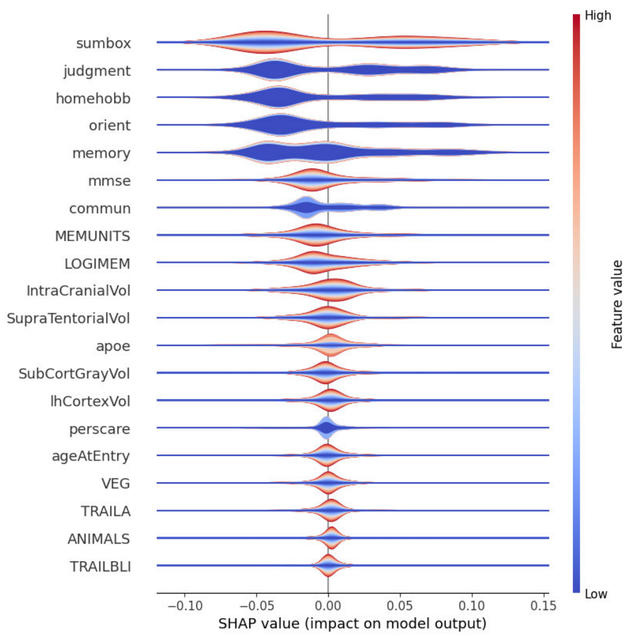

FIGURE 6. **SHAP violin plot for AD class.**

feature values for each data point. This color gradient makes it easier to see how a feature's value influences its SHAP value. For example, high feature values (red) may systematically lead to positive SHAP values, indicating that the feature influences predictions favorably when it is high. The features near the peak of the violin plot are more relevant to the model's estimations, while those at the bottom are less significant. The width and spread of the violin represent the heterogeneity in how each attribute affects predictions. The import features can be easily identified by the long tails on the right side of the x-axis.

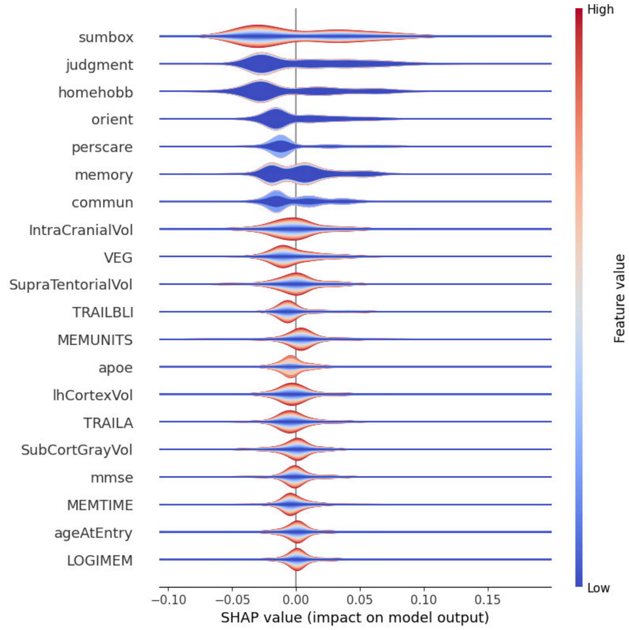

FIGURE 7. **SHAP violin plot for Non-AD class.**

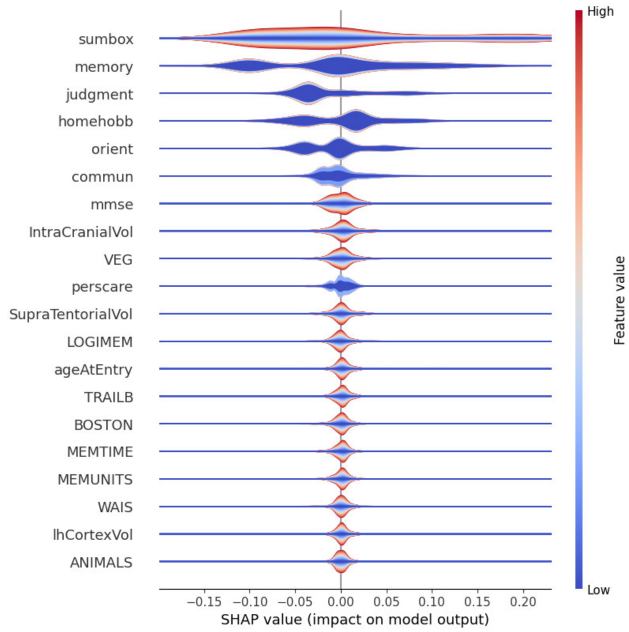

FIGURE 8. **SHAP violin plot for uncertain class.**

From Figure 5, it is observed that memory, sumbox (0), judgment, commun, and orient are the top five essential features. The low memory, sumbox, judgment, commun, and orient values are responsible for CN prediction.

Furthermore, sumbox is the most important feature for predicting AD Figure 6. Here, the higher value of sumbox (mostly 2-5) is influencing positively. Similarly, a bit higher value (mostly 1) of judgment, homehobb, orient, and memory positively influence AD identification. Figure 7 illustrates

TABLE 8. Top five influential features of each class.

Class	Features	Sumbox Value
CN	Memory, sumbox, judgment, commun, and orient	0
AD	Sumbox, judgment, homehobb, orient, and memory	2-5
Non-AD	Sumbox, judgment, homehobb, orient, and perscare	5-10
Uncertain	Sumbox, memory, judgment, homehobb, and orient	0.5-2.5
Others	Sumbox, judgment, memory, intraCranialVolume, and homehobb	0, 0.5, 0.75, and 1

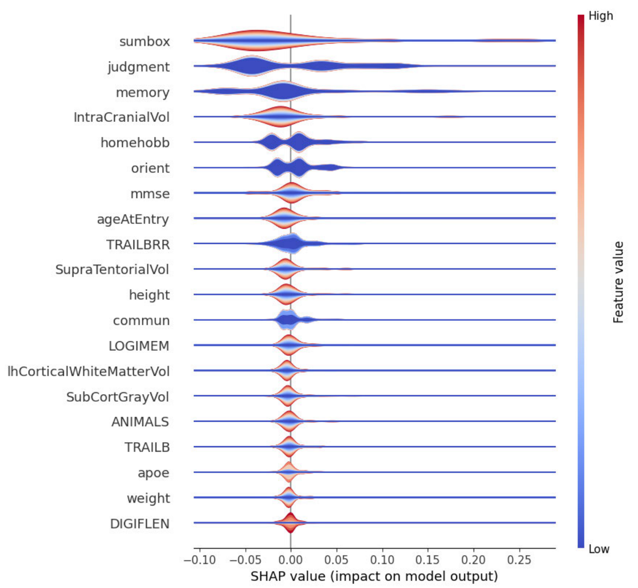

FIGURE 9. SHAP violin plot for others class.

that the very high value of sumbox (mostly 5-10) is responsible for Non-AD prediction. For Non-AD prediction judgment, homehobb, orient, perscare, and memory are the other most important features, and their value lies between (0.5-2). Afterward, sumbox (mostly 0.5-2.5), memory, judgment, homehobb, and orient are the most influential features for classifying the uncertain class, which is represented in Figure 8. Finally, for others class in Figure 9, sumbox (mostly 0, 0.5, 0.75, and 1), judgment, memory, intraCranialVolume, and homehobb are the top five most influential features. As the sumbox is the most common and influential feature in each class, the value range of the sumbox is also stated here. Table 8 expresses the top influential features sequentially.

VI. CONCLUSION

This paper proposes a Federated XAI-based framework that preserves privacy to predict AD using multimodal data. The use of FL guarantees that sensitive patient data remains localized, protecting privacy while allowing collaboration between many institutions. The model incorporates several components of Alzheimer's progression by combining multimodal data, such as MRI scans, clinical data, and psychological data, collected from the OASIS-3 repository, resulting in more accurate and comprehensive predictions. The introduction of XAI improves the predictions of the model by providing insight into the crucial factors driving disease progression, which is critical for both physicians and researchers. This interpretability builds trust in AI-driven forecasts, enabling healthcare professionals to make better judgments. In general, this method shows that FL, when combined with explainability and multimodal data, is a promising way to improve AD prediction, improve patient outcomes, and protect data privacy. The model's accuracy, precision, recall, F1-score, and AUC are 98.93%, 98.94%, 98.93%, and 99.97%, respectively. Future research can concentrate on refining the model's performance across several additional data modalities to improve predicted accuracy. As this work focused on the FL model rather than the system, system performance metrics are not calculated as performance metrics. Therefore, system performance metrics, such as training time, computation cost, communication overhead, and latency, could be measured in the future when establishing a complete FL system.

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